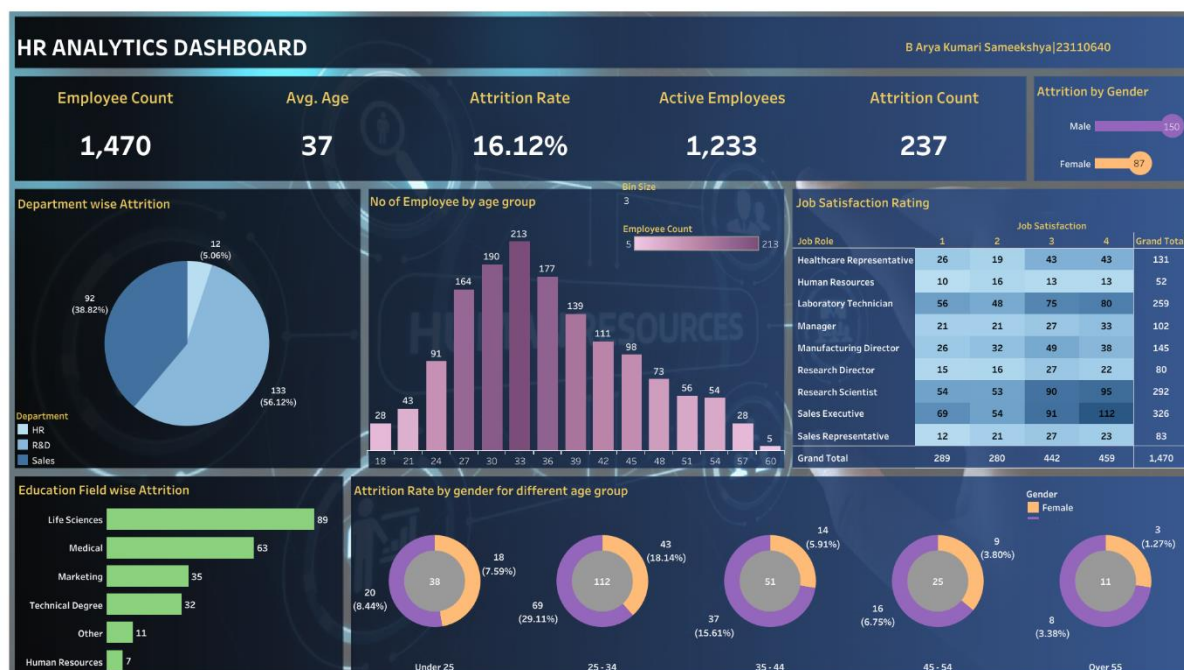
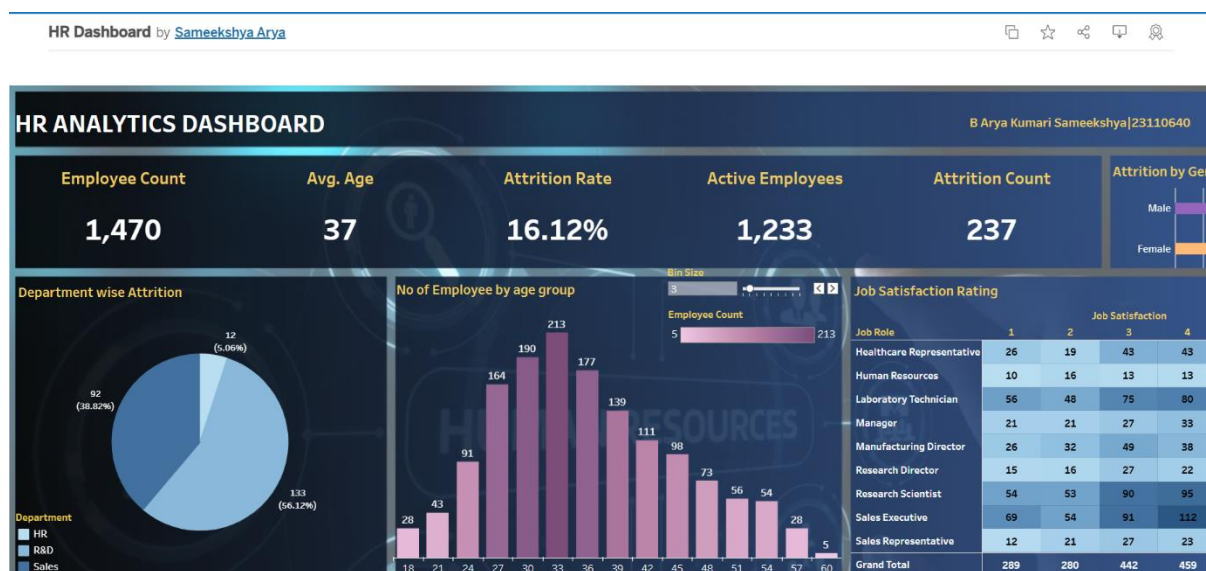


Data and Visual Analytics in AI

Assignment-2

1) Create a dashboard as shown in the figure using Tableau.



Link to my dashboard:

https://public.tableau.com/app/profile/sameekshya.arya/viz/HRDashboard_17621131998520/HRAnalyticsDashboard?publish=yes

- 2) Provide a case study AI application in any one of the following- Healthcare, Finance, Retail, Supply Chain Management, Defence. Mark weightage will be given depending on depth of analysis.

Case Study: Artificial Intelligence in Defence — AI-Powered Drone Surveillance and Threat Detection

1. Introduction

In modern warfare, the speed and accuracy of decision-making are critical. With the rise of complex battlefields, unmanned systems, and hybrid warfare, Artificial Intelligence (AI) has become an indispensable tool for defense organizations worldwide. One of the most impactful applications of AI in defence is AI-powered drone surveillance and autonomous threat detection — enabling real-time intelligence, precision targeting, and enhanced situational awareness with minimal human risk.

2. Problem Statement

Traditional surveillance and reconnaissance operations rely heavily on manual image analysis and human-operated drones, which are time-consuming and prone to human fatigue or oversight. In large and complex terrains (like border areas or conflict zones), continuously monitoring video feeds for potential threats such as enemy movements, vehicles, or weapons is nearly impossible for human operators alone.

Challenge:

How can we use AI to automate surveillance, detect potential threats in real-time, and assist military personnel in rapid, accurate decision-making?

3. AI Solution Overview

Core Concept:

Integrate Computer Vision and Deep Learning algorithms into unmanned aerial vehicles (UAVs)** to automatically detect, classify, and track objects of interest (e.g., enemy vehicles, soldiers, weapons, or drones) in real-time video feeds.

Key Technologies Used

- **Machine Learning (ML):** To train models on thousands of labeled battlefield images.
- **Computer Vision:** For object recognition, motion detection, and target tracking.

- **Deep Learning (CNNs):** For analyzing live video streams and identifying threats.
 - **Reinforcement Learning:** For autonomous decision-making and path optimization.
 - **Edge AI:** Processing data on the drone itself (onboard chips) to reduce latency.
-

4. Workflow of the AI System

1. **Data Collection:** Satellite images, aerial surveillance videos, and radar data are gathered.
 2. **Model Training:** Deep neural networks are trained to recognize specific targets (e.g., tanks, vehicles, human groups, missile launchers).
 3. **Real-Time Processing:** The drone captures live footage and processes it using AI chips onboard (like NVIDIA Jetson).
 4. **Threat Detection:** The system flags suspicious activities or objects and sends alerts to command centers.
 5. **Decision Support:** AI suggests next actions (e.g., track, zoom, deploy defense, or ignore false alarms).
-

5. Real-World Examples

A. Project Maven (USA)

- Developed by: U.S. Department of Defense (DoD).
- Purpose: Analyze hours of drone footage to identify vehicles, people, and threats automatically.
- Technology: TensorFlow-based deep learning models trained on classified datasets.
- Impact: Reduced human workload by 80% in video analysis and increased accuracy in target identification.

B. DRDO's AI-Based Surveillance in India

- Developed by: Defence Research and Development Organisation (DRDO), India.
- Use Case: AI-powered systems for border surveillance using drones and ground sensors.
- Features:
 - Intrusion detection through motion recognition.
 - Real-time alerts to army control centers.
 - Night vision and thermal imaging integration.

- Impact: Enhanced security at high-risk borders like LOC and LAC with reduced manpower requirements.

C. Israel’s Harpy and Harop Drones

- Use: AI-driven “loitering munitions” that autonomously identify and destroy radar emitters.
 - Feature: Capable of operating without real-time human control once launched.
-

6. Results and Impact

Parameter	Traditional Surveillance	AI-Powered System
Human Involvement	High	Minimal
Detection Speed	Slow (minutes–hours)	Real-time (< 2 sec)
Accuracy	70–80%	90–98%
Risk to Soldiers	High	Significantly Reduced
Coverage	Limited to operator’s capacity	24x7 Autonomous

Overall Impact:

- Enhanced national security through continuous border monitoring.
 - Faster threat response and reduced operational delays.
 - Cost-effective compared to large-scale human surveillance operations.
 - Minimized risk to soldiers in dangerous zones.
-

7. Challenges and Limitations

- Ethical Concerns: Risk of autonomous lethal decisions without human oversight.
 - Data Security: AI systems must resist cyberattacks and data breaches.
 - False Positives: Misidentification of non-hostile targets could lead to civilian harm.
 - Dependence on Data Quality: Poor-quality or biased data can affect accuracy.
 - International Regulation: AI weapons are still under global debate for control and legality.
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8. Future Scope

- Integration with 5G: Enables faster, secure communication between drones and control centers.
 - Swarm Intelligence: Multiple AI drones coordinating autonomously in missions.
 - Predictive Defence Systems: AI models predicting attacks before they occur using pattern analysis.
 - AI-Driven Cybersecurity: Protecting military networks using anomaly detection algorithms.
 - Human-AI Collaboration: Keeping a “human in the loop” for ethical decision-making.
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9. Conclusion

AI has revolutionized defence by making surveillance, reconnaissance, and threat detection more intelligent, accurate, and autonomous. Through projects like Project Maven and DRDO’s AI surveillance drones, nations are strengthening their security while reducing human risk. However, balancing autonomy with ethics and innovation with control remains crucial as AI continues to reshape the future of warfare.

10. References

1. U.S. Department of Defense – *Algorithmic Warfare Cross-Functional Team (Project Maven)*, 2017.
2. DRDO Annual Report 2023 – *AI Applications in Defence Surveillance Systems*.
3. NATO Review (2022) – *The Future of Artificial Intelligence in Military Applications*.
4. Indian Express (2024) – *AI in Defence: DRDO’s Surveillance Drones for Border Security*.